

TESLA STOCK DATA

Last updated by IIC - Jun 17, 2020

Number of Columns - 7

DESCRIPTION:

Tesla, Inc. is an American electric vehicle and clean energy company based in Palo Alto, California. Tesla's current products include electric cars, battery energy storage from home to grid-scale, solar products, and related products and services.

Here is the Latest stock data of Tesla till 2020.

Attributes

- *Date*
- *Open*
- *High*
- *Low*
- *Close*
- *Adj Close*
- *Volume*

DATASET LINK:

<https://www.kaggle.com/salehahmedrony/tesla-latest-stock-data-2010-2020?select=TSLA.csv>

Libraries / Platforms / Tools

1. *numpy*
2. *pandas*
3. *matplotlib*
4. *Google Colab*

Questions

1. Write a Python program to load the dataset in a *pandas* **DataFrame** and display the number of rows and columns in the following format.

```
(num_of_rows, num_of_cols)
```

2. Display the first 5 rows and last 10 rows of the dataframe created in (1) using *pandas*.
3. Write a Python program to list the data types of all attributes in the dataset, each in a new line, in the following output format.

attribute_name - data_type

An example output can be the following.

Date	object
Open	float64
High	float64

... and so on.

4. Write a Python program to display the maximum and minimum values of all numerical attributes in the dataset in the following format.

	max	min
Open	5.021400e+02	3.228
High	5.024900e+02	3.326
Low	4.705100e+02	2.996

... and so on.

5. Write a Python program to find the count of missing values present in each attribute and print it in the following format.

attribute_name - count_of_missing_values

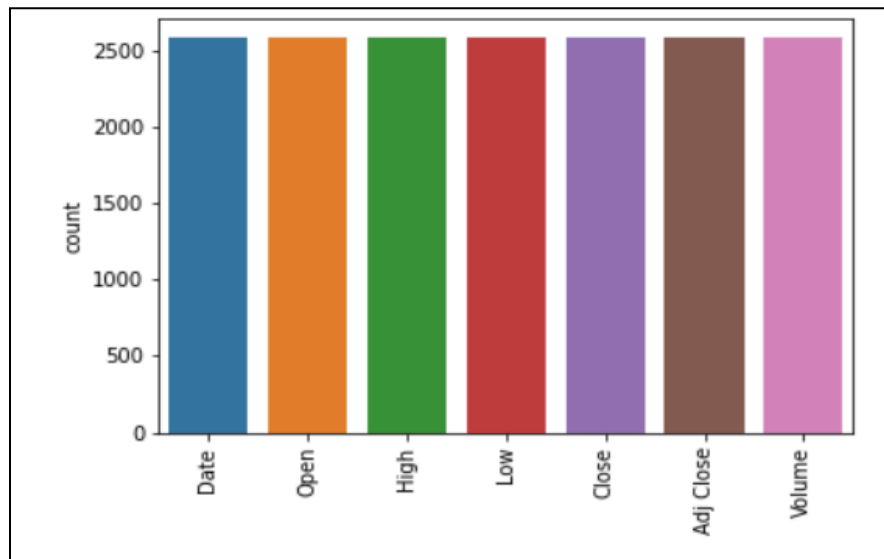
An example output can be the following.

Date	0
Open	0
High	0

... and so on.

6. Draw a **Count Plot** to show the number of missing values graphically.

HINT- <https://seaborn.pydata.org/generated/seaborn.countplot.html>



7. Write a Python program to calculate the following for all numerical attributes.
- Mean
 - Standard deviation
 - Median

Display the output in the following format.

	Open	High
mean	49.206686	50.301806
std	57.934102	59.888383
median	44.001999	44.660000

.. and so on (for the rest of the attributes)

HINT -

https://pandas.pydata.org/pandas-docs/stable/getting_started/intro_tutorials/06_calculate_statistics.html

8. Write a Python program to display a heatmap of the **Correlation Matrix** between all the numerical attributes using the *seaborn* library.



HINT -

<https://medium.com/@szabo.bibor/how-to-create-a-seaborn-correlation-heatmap-in-python-834c0686b88e>

9. Write a Python program to change the data type of **DATE** attribute to date-time datatype.

Date	datetime64[ns]
Open	float64
High	float64
Low	float64
Close	float64

10. Write a Python program to display the values of **Date(s)** and **Open** of the stock where **High** is greater than 450 as follows.

	Date	Open
0	2020-08-27	436.092010
1	2020-08-28	459.023987
2	2020-08-31	444.609985
3	2020-09-01	502.140015
4	2020-09-02	478.989990
5	2020-09-15	436.559998
6	2020-09-16	439.869995
7	2020-09-18	447.940002
8	2020-09-21	453.130005

HINT - <https://www.analyticsvidhya.com/blog/2020/02/loc-iloc-pandas/>

11. Write a Python program to store the top 10 values of **Date(s)** and **High** with respect to **Close** prices in a DataFrame. Display the dataframe as follows.

	Date	High
0	2020-08-31	500.140015
1	2020-09-01	502.489990
2	2020-09-15	461.940002
3	2020-09-21	455.679993
4	2020-08-27	459.119995
5	2020-09-02	479.040009
6	2020-08-28	463.697998
7	2020-09-18	451.000000
8	2020-09-16	457.790009
9	2020-08-26	433.200012

HINT - <https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.nlargest.html>

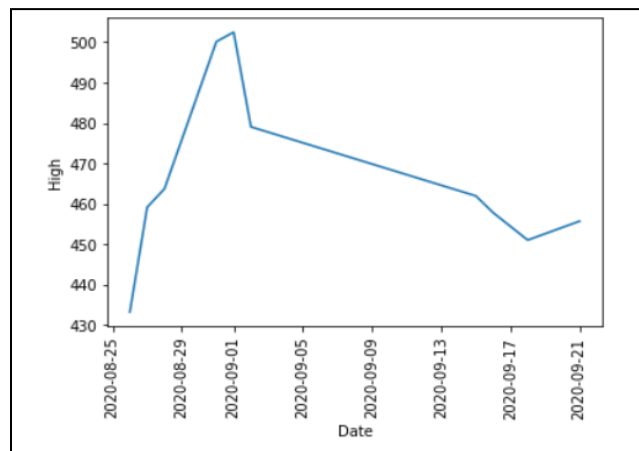
12. Write a Python program to store the bottom 10 values of **Date(s)** and **Low** with respect to **Open** prices in a DataFrame. Display the dataframe as follows.

	Date	Low
0	2010-07-08	3.114
1	2010-07-07	2.996
2	2010-07-13	3.380
3	2010-07-09	3.310
4	2010-08-12	3.478
5	2010-07-14	3.552
6	2010-07-12	3.400
7	2010-08-13	3.532
8	2010-08-16	3.652
9	2010-08-19	3.666

HINT -

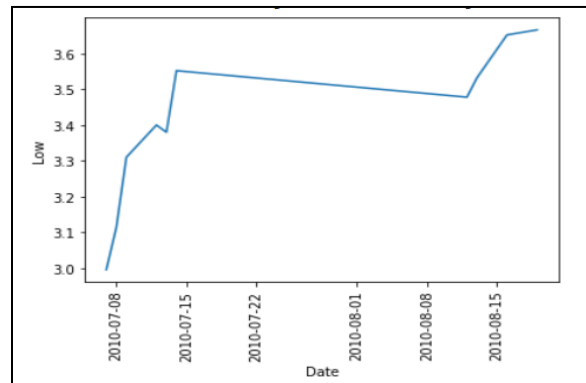
<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.nsmallest.html>

13. Write a Python program to draw a line plot of **Date** versus **High** values in the dataframe created in (11).



HINT - <https://seaborn.pydata.org/generated/seaborn.lineplot.html>

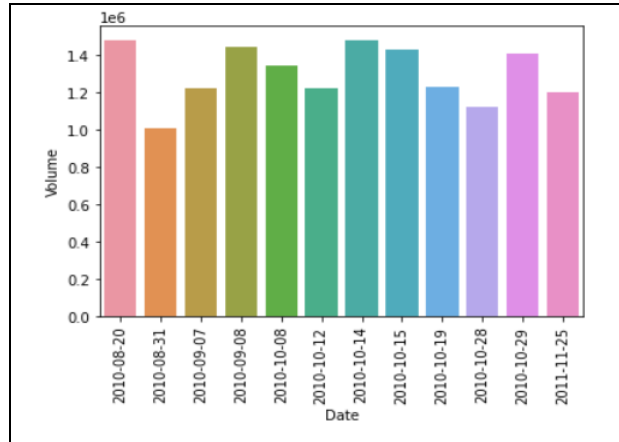
14. Write a Python program to draw a line plot of **Date** versus **Low** values in the dataframe created in (12).



15. Write a Python program to store the values of **Date(s)** and **Volume** attributes in a dataframe where the **Volume** of shares traded is between 1000000 and 1500000. Display the dataframe as follows.

	Date	Volume
0	2010-08-20	1480000
1	2010-08-31	1005500
2	2010-09-07	1217000
3	2010-09-08	1442000
4	2010-10-08	1339000
5	2010-10-12	1220000
6	2010-10-14	1474000
7	2010-10-15	1423500
8	2010-10-19	1226000
9	2010-10-28	1121000
10	2010-10-29	1403000
11	2011-11-25	1198000

16. Write a Python program to draw a bar plot of **Date** versus **Volume** values using the dataframe created in (15) as depicted below.



HINT :- <https://seaborn.pydata.org/generated/seaborn.barplot.html>

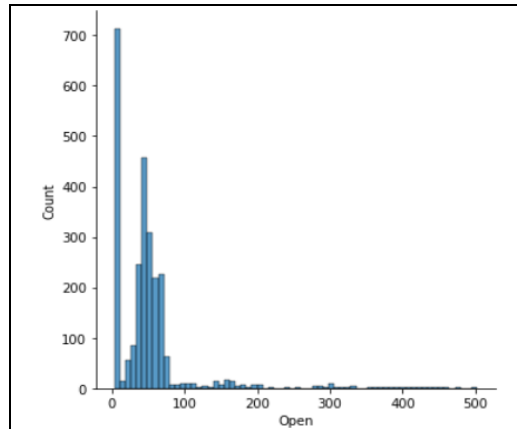
17. Write a Python program to display the values of **Date**, **High** and **Low** where **Open** is equal to **Close** as follows.

	Date	High	Low
0	2010-10-08	4.158000	4.078
1	2011-02-11	4.750000	4.588
2	2012-03-07	6.662000	6.582
3	2012-04-24	6.440000	6.200
4	2012-07-06	6.346000	6.160
5	2019-12-17	77.099998	75.180

18. Write a Python program to plot the distributions of values in the form of histogram for the following attributes as shown in the example below.

- Open**
- High**
- Low**
- Close**
- Volume**

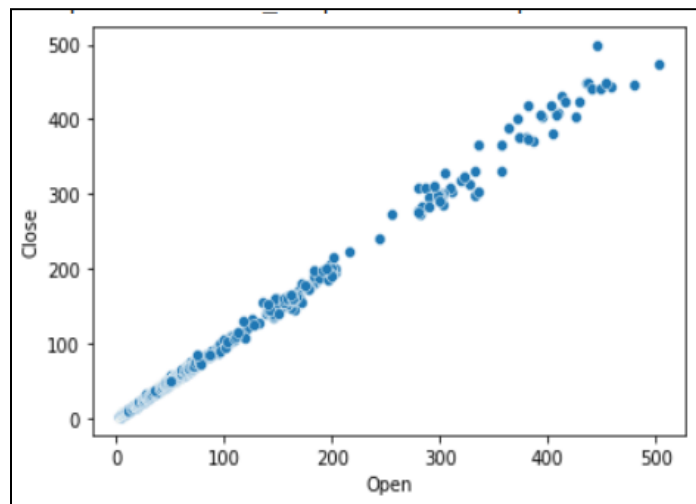
Example of a distribution using histogram



19. Write a Python Program to plot the **Scatter Plot** between the following attributes.

- a. **Open vs Close**
- b. **High vs Low**
- c. **Volume vs Adj Close**

Example of a Scatter-plot



HINT - <https://seaborn.pydata.org/generated/seaborn.scatterplot.html>

Answer -

<https://colab.research.google.com/drive/1UQZTMj2zxC1VmxwgZEkJEIXv8XvhpvHk?usp=sharing>